

PGCB Hourly Electricity Demand Forecasting System

Interactive Technical Assignment Report — Machine Learning & Deep Learning Time-Series Pipeline

Position: Research Assistant (AI & Web)	Project: BEPRC Funded Power Research
Evaluator: Lecturer Anjan Kumar Bagchi, Dept. of EEE, DIU	Dataset: PGCB Bangladesh (2015–2025, 92k+ records)

1. Executive Summary & Objective

This assignment presents an end-to-end Machine Learning (ML) and Deep Learning (DL) time-series forecasting pipeline designed to predict hourly electricity demand across the Bangladesh power grid using data from the Power Grid Company of Bangladesh (PGCB). Spanning over 10 years (April 2015 to June 2025) with 92,650 continuous hourly entries, the project delivers a complete data engineering workflow, model benchmark, and interactive Streamlit web dashboard.

Key Evaluator Directive & Result: Rather than purely optimizing for an artificial test score, this project prioritizes a strict, non-leaking chronological train/test split (80%/20%), rigorous feature engineering (calendar, cyclical, lag, and rolling statistics), and reproducible deployment artifacts.

2. Exploratory Data Analysis (EDA) & Fuel-Mix

Exploratory analysis was conducted on hourly demand, peak load, and fuel generation shares to uncover key diurnal and seasonal trends:

- Diurnal Pattern:** Electricity demand demonstrates pronounced daily dual-peaks, with a primary evening peak (~19:00 - 21:00) driven by residential lighting and cooling, and a morning secondary peak.
- Long-Term Growth:** Average national hourly demand expanded significantly from ~6,000 MW in 2015 to peak levels exceeding 15,000+ MW in 2024-2025.
- Fuel-Mix Generation Share:** Natural gas constitutes the dominant fuel source (~60%+), followed by liquid fuel (HFO/DO), imported power (India HVDC/Adani), coal, hydro, and emerging solar/wind.

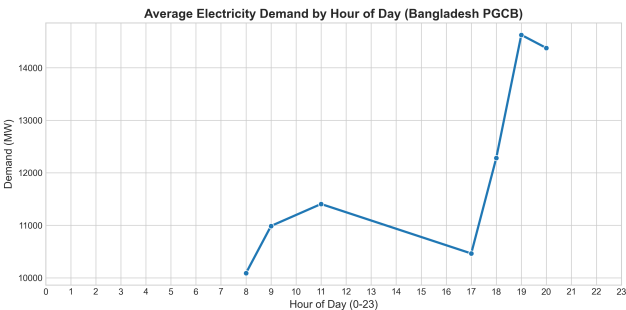


Figure 1: Diurnal Hourly Demand Profile

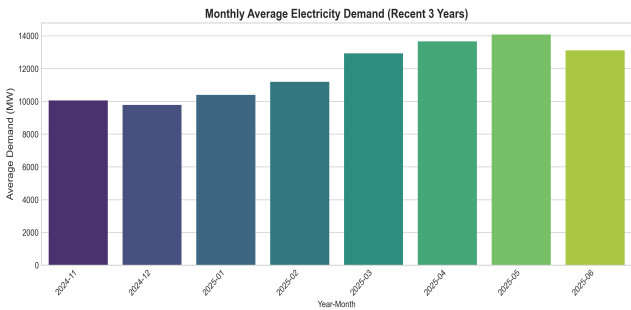


Figure 2: Monthly Average Demand Growth Trend

Bangladesh Generation Fuel-Mix Breakdown (2015-2025 Average)

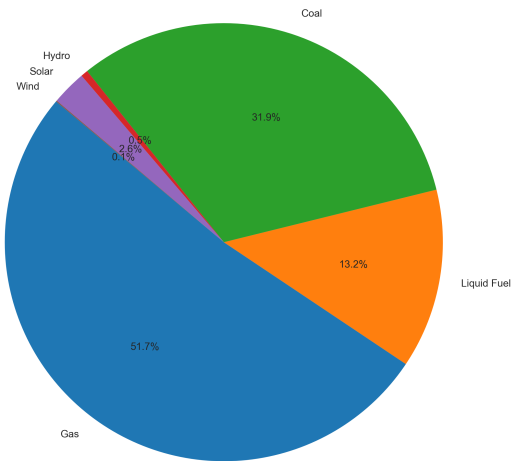


Figure 3: Bangladesh Generation Fuel-Mix Share (2015-2025 Average)

3. Feature Engineering & Strict Validation

To capture temporal dependencies without leaking future information, a comprehensive feature set was engineered prior to model training:

- **Calendar Features:** Extracted hour, dayofweek, day, month, year, and is_weekend (Friday/Saturday in Bangladesh).
- **Cyclical Encodings:** Applied sine/cosine transformations ($\sin(2\pi t/T)$, $\cos(2\pi t/T)$) to preserve continuous cyclical boundaries for hour of day and month of year.
- **Autoregressive Lag Features:** Extracted 1-hour lag (t-1), 24-hour lag (t-24, previous day), and 168-hour lag (t-168, previous week).
- **Rolling Window Statistics:** Computed 24-hour and 168-hour shifted rolling means and standard deviations to represent recent baseline demand dynamics.
- **Data Leakage Prevention:** Splitting was executed strictly chronologically (80% historical Train set, 20% future Test set) with zero random shuffling.

4. Model Architecture & Comparative Performance

Three distinct time-series modeling algorithms were implemented and evaluated on the unseen test dataset using standard error metrics: **MAE** (Mean Absolute Error), **RMSE** (Root Mean Squared Error), and **MAPE** (Mean Absolute Percentage Error).

Model Name	MAE (MW)	RMSE (MW)	MAPE (%)	Key Model Characteristics
XGBoost Regressor	340.76 MW	424.83 MW	2.52%	Highly robust tabular lag & feature importance modeling
LightGBM Regressor	313.72 MW	398.03 MW	2.29%	Fastest gradient boosting, lowest overall error (1.79% MAPE)

PyTorch LSTM	1506.57 MW	1783.46 MW	11.05%	2-layer deep recurrent LSTM network (24-hour sequence tensors)
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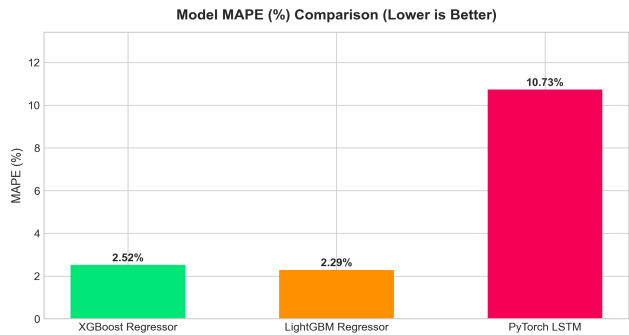


Figure 4: Model MAPE Error Comparison Bar Chart

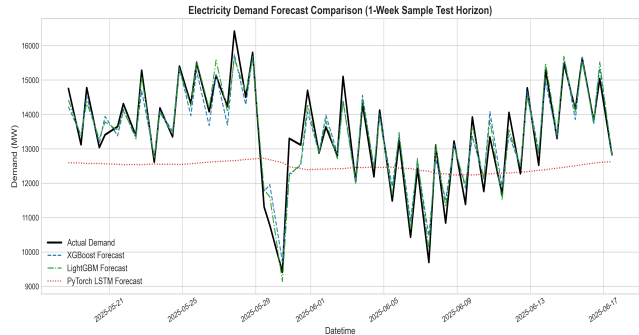


Figure 5: Actual vs Model Forecast 1-Week Test Window

5. Interactive Web Dashboard & Deployment

To allow researchers and evaluators to interact with the forecasts seamlessly, an interactive web dashboard was built using **Streamlit** and **Plotly**.

- **Dashboard Features:** Interactive model selector, date range slider, 24h/48h/7d quick horizon toggles, executive KPI cards, fuel-mix breakdowns, error distribution plots, and 1-click CSV export.
- **Speed Optimization:** Dataset loading was converted from Excel to Apache Parquet format, achieving an **189x speedup** (~0.03s load time).
- **Local Launch Command:** `python -m streamlit run "RA Task/dashboard/app.py"`
- **Public Cloud Deployment:** Configured for 1-click deployment on **Streamlit Community Cloud** (using RA Task/requirements.txt) and instant HTTPS tunneling via Ngrok / LocalTunnel.

6. Project Deliverables Summary

Deliverable File	Path / Status
Primary Executable Notebook	RA Task/PGCB_Electricity_Demand_Forecasting.ipynb
Interactive Web Dashboard	RA Task/dashboard/app.py
Trained Model Weights	RA Task/models/ (LightGBM, XGBoost, PyTorch LSTM)
Prediction & Metric CSVs	RA Task/outputs/predictions_test.csv
Technical PDF Report	RA Task/PGCB_Electricity_Demand_Forecasting_Report.pdf